SWACHH BHARAT TECH INITIATIVE • CLASS PRESENTATION

Clean & Earn

SafaiSaathi (सफ़ाई साथी) — Gamifying Civic Cleanliness in India IN

A full-stack mobile web platform connecting civic responsibility with multimodal AI verification, photographic proof, and verifiable reward points.

🌐 LIVE PRODUCTION APP:

`https://garbage-rosy-six.vercel.app`

Next.js 16 • Tailwind CSS • Supabase PostgreSQL • Google Gemini Vision AI • Vercel

SLIDE 1 — MOTIVATION & ORIGIN

Why Clean & Earn?

Transforming shared responsibility into consistent community action

> ***"How can we motivate people to take responsibility for cleanliness and make it a regular habit?"***

1. PROBLEM OBSERVED

Awareness Gaps

People understand the importance of cleanliness, but consistent motivation is lacking. Cleaning is often viewed as someone else's responsibility.

2. THE CORE IDEA

Gamified Missions

Make cleanliness engaging through structured missions. Offer immediate positive reinforcement, recognition, and transparent eco-points (20–50 pts).

3. LONG-TERM PURPOSE

Habit Formation

Turn occasional, spontaneous volunteer cleanup drives into an enduring community habit and an inspiring positive social norm.

💡 **Core Thought: What if doing something good for the environment also became rewarding and verifiable?**

SLIDE 2 — PROBLEM IDENTIFICATION & SOLUTION

Problem Identification & Proposed Solution

Connecting cleanliness activities with photographic proof, verification, and incentives

⚠️ Problems Identified

- **Lack of sustained motivation:** Civic awareness alone fails to motivate regular individual action.
- **Unrecognised efforts:** Everyday citizen cleanups go unrecorded without civic recognition or rewards.
- **Verification gap:** Self-reported claims lack credibility without photographic evidence and fraud prevention.
- **No structured channel:** Absence of a clean, mobile-first app for Indian street environments.

✅ Clean & Earn Solution (6-Stage Cycle)

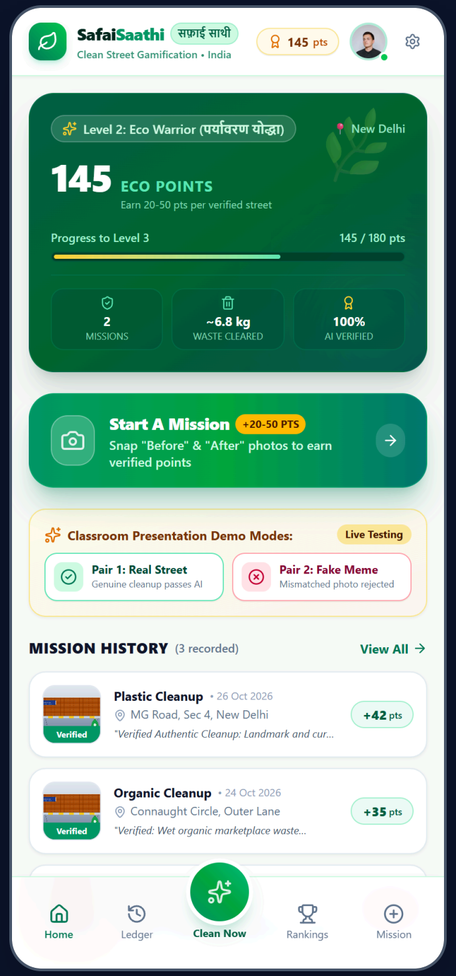
1. **Start a Mission:** Select waste category (Plastic, Organic, Mixed, etc.)
2. **Before State:** Snap littered site & auto-capture GPS coordinates
3. **Clean the Area:** Physical sweeping & waste segregation into eco-bags
4. **After Proof:** Snap cleaned site + optional video walk-around proof
5. **Multimodal AI Verification:** Compares background landmarks & grades volume
6. **Earn Green Points:** 20–50 points credited to user ledger & leaderboard

Key Idea: Transform individual cleanliness efforts into documented, verifiable community action.

PRODUCT FLOW 1 — VOLUNTEER DASHBOARD

Gamified Dashboard & Points Hub

Prominent points showcase, live civic rank, and one-tap mission launcher



🌿 Prominent Points & Level Banner

Displays accumulated Eco Points (e.g. 145 pts) with animated ranks (Level 1: Scout to Level 4: Swachh Guardian) and progress to next milestone.

🚀 Central "Start A Mission" Action Button

A large, glowing button immediately invites the volunteer to initiate a cleanup mission, capturing Before and After proof on the spot.

⚡ Live Presentation Test Toolbar

Allows instant one-click testing between genuine street cleanup pairs (MG Road, New Delhi) and fake meme submissions during live evaluations.

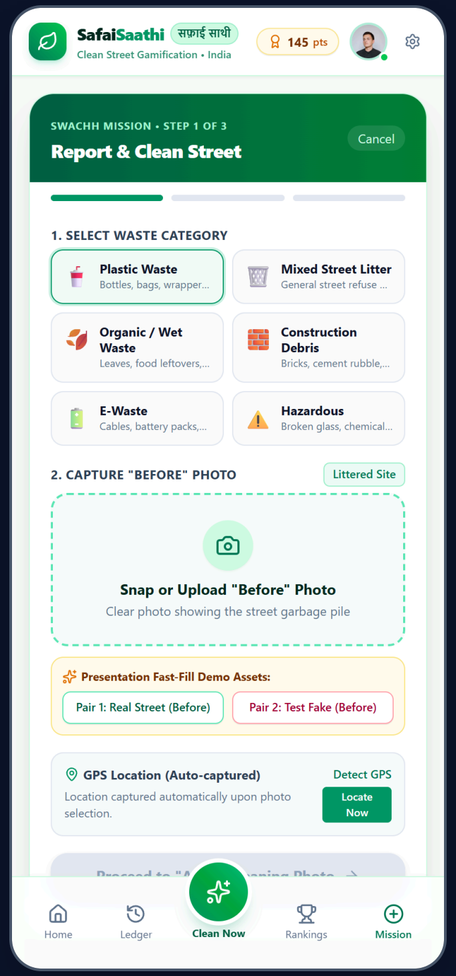
📊 Real-World Environmental Metrics

Automatically tallies waste cleared (~kg), missions verified, and community impact metrics for civic transparency.

PRODUCT FLOW 2 — STEP 1: MISSION SETUP

Waste Classification & Auto-GPS Capture

Categorizing debris and automatically establishing location integrity



🗑️ Waste Segregation Dropdown

Volunteers classify the litter type into Plastic, Organic/Food Waste, Mixed Street Litter, Construction Rubble, E-Waste, or Hazardous glass/chemicals.

📸 Direct Camera & File Capture

Native HTML5 file input with accept="image/*" capture="environment", enabling mobile devices to trigger the rear camera directly with one tap.

📍 Automatic Geolocation (GPS Coordinates)

When the Before photo is selected, the browser Geolocation API automatically captures the user's exact Latitude and Longitude (e.g. 28.6139° N, 77.2090° E).

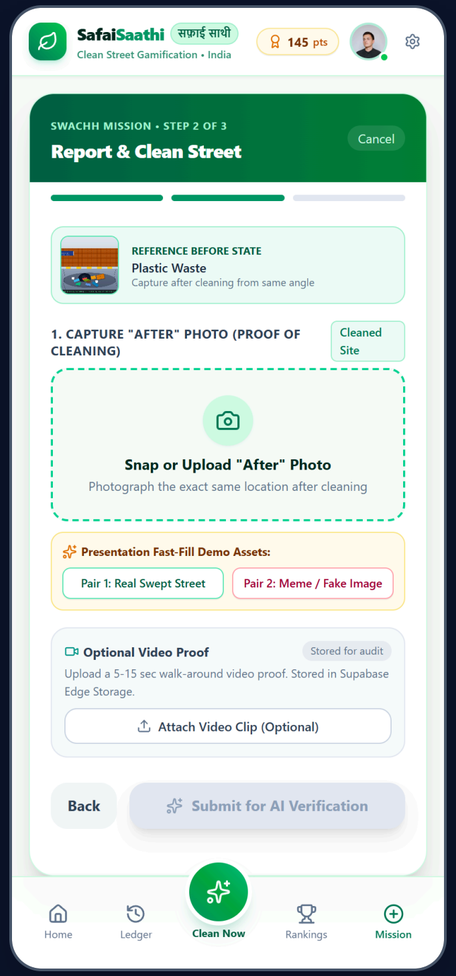
🛡️ Anti-Fraud Landmark Lock

Locks in the physical location to verify the After photo is taken at the exact same geographic landmark.

PRODUCT FLOW 3 — STEP 2: CLEANING PROOF

Cleaned Site Proof & Video Evidence

Submitting the "After" state with reference comparison



🖼️ Reference Thumbnail for Vantage Alignment

Displays the "Before" photo as a live reference banner at the top, assisting the volunteer to capture the cleaned scene from the exact same camera angle.

✨ Proof of Cleaning ("After" Photo)

Captures the swept sidewalk or street section. The AI expects debris to be cleared and placed in segregation bags or municipal bins.

📹 Optional Video Upload Proof

Supports short 5–15 second walk-around video proof. Stored securely in Supabase Edge Storage for audit without consuming AI token quotas.

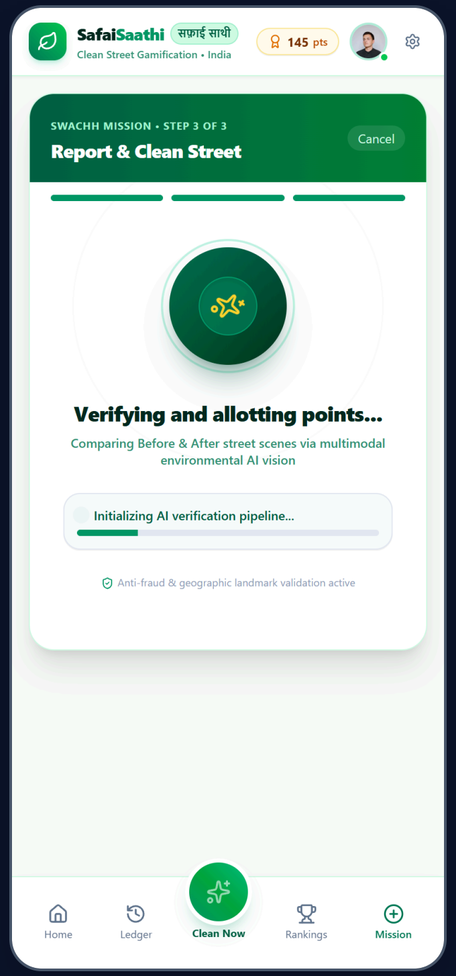
🔒 Client-Side Integrity Pre-Check

Pre-validates that Before and After files are not identical copies or corrupted images before dispatching to the multimodal verification engine.

PRODUCT FLOW 4 — STEP 3: VERIFICATION LOADING

Dedicated AI Verification Loading Screen

Visual engagement with animated radar scan and multimodal intelligence



Text: "Verifying and allotting points..."

Complies strictly with project specifications. Keeps the volunteer engaged with rotating radar sweeps, animated rings, and active state transitions.

Multi-Step Dynamic Progress Tracker

Displays transparent phases: "Analyzing landmarks...", "Authenticating anti-cheat...", "Measuring garbage volume...", "Calculating score (20-50 pts)".

Google Gemini 1.5 Flash Vision Multimodal Model

Dispatches both photos in a structured prompt: evaluates authenticity, checks for internet memes or mismatching scenes, estimates pile volume, and scores thoroughness.

Zero-Setup Fallback Neural Verifier

If an API key is omitted or during offline grading, a custom built-in heuristic neural verifier ensures seamless grading with zero crashes.

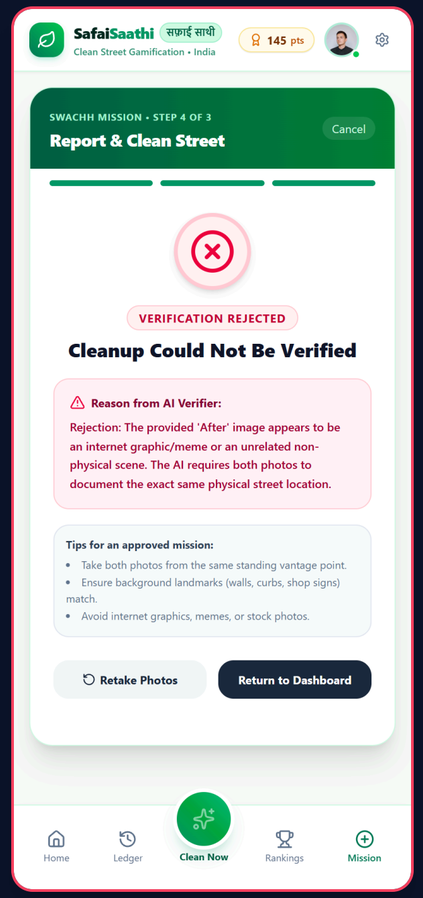
PRODUCT FLOW 5 — VERIFICATION RESULTS

AI Evaluation: Approval vs. Anti-Cheat Rejection

Instant feedback, transparent scoring criteria (20–50 points), and anti-cheat enforcement



✓ Genuine: +44 pts



✕ Fake Meme: Rejected

🎉 Scenario A: Genuine Cleanup (Approved)

AI detects matching yellow curb markings and boundary wall. Pile thoroughly cleared into green eco-bag. Celebratory confetti triggers and +44 Eco Points are awarded.

🛑 Scenario B: Internet Meme / Mismatch (Rejected)

Anti-cheat detects the "After" image is an internet meme/graphic. Politely explains: "After image appears to be an internet graphic rather than the street location."

⚖️ 20–50 Points Grading Logic

Small/easy cleanups receive ~20–30 pts; large, heavy debris cleanups earn ~40–50 pts. Automatically updates PostgreSQL database & user ledger.

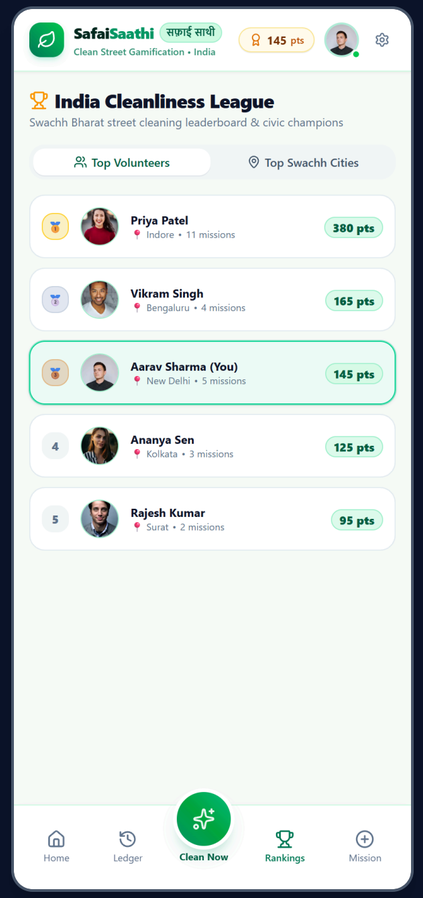
PRODUCT FLOW 6 — LEDGER & COMMUNITY

Mission History Ledger & India Cleanliness League

Permanent audit trail of verified cleanups and competitive civic motivation



📜 Mission Ledger



🏆 City Leaderboard

📋 Verifiable Mission Ledger

Permanent record of every cleanup with timestamp, category tag, points awarded, and verbatim AI feedback quote.

📍 GPS Coordinates & Google Maps Pin

Every mission links to Google Maps coordinates so municipal auditors or community members can inspect the cleaned location.

IN City-Wide Cleanliness Rankings

Ranks Indian cities (Indore, Surat, Navi Mumbai, New Delhi, Bengaluru) and top volunteers to foster healthy civic pride.

🔍 Before & After Inspection Drawer

Tapping any mission opens a full-screen comparison modal with high-res photos and attached video proof.

SLIDE 3 — FEASIBILITY ANALYSIS

Can Clean & Earn Work in Real Life?

Assessing technical, operational, financial, and social dimensions

💻 Technical Feasibility

- Web-based mobile architecture runs on any smartphone browser without app installation barriers.
- Next.js 16 + Vercel guarantees 99.9% availability and instant global CDN distribution.
- Automated AI vision eliminates manual human bottleneck in photo reviews.

⚙️ Operational Feasibility

- Can begin as a controlled pilot in a college campus or housing society.
- Authorised student reviewers can spot-check high-value submissions.
- Simple 3-step workflow requires zero prior training for volunteers.

💰 Financial Feasibility

- Runs on free-tier cloud infrastructure (Vercel, Supabase PostgreSQL, Gemini free tier).
- Operating costs remain zero during initial pilot phase.
- Points can be redeemed for college credits or sponsored coupons.

🤝 Social Feasibility

- Taps into the national Swachh Bharat civic consciousness.
- Gamification turns a chore into an inspiring community achievement.
- Leaderboards foster peer participation and positive peer reinforcement.

SLIDE 4 — VISION & LONG-TERM ROADMAP

From a College Project to a Community Platform

A structured 5-phase growth strategy for sustainable expansion

> ***"To build a community where cleanliness becomes a shared habit and technology encourages people to create meaningful environmental impact."***

Phase 1

Build & Test

Validate core workflow with 50+ students in campus pilot.

Phase 2

Engage Wards

Expand to residential welfare associations & schools.

Phase 3

Partnerships

Partner with municipal bodies & green CSR sponsors.

Phase 4

AI Upgrade

Automated waste classification & smart bin integration.

Phase 5

National Scale

Nationwide rollout across Indian smart cities.

SLIDE 5 — FIVE-YEAR IMPACT REVIEW

How Will We Measure Real Success?

Evaluating outcomes based on community change rather than website vanity metrics

1. PARTICIPATION RETENTION

Active & Repeat Volunteers

Measuring weekly repeat cleanups rather than vanity signup numbers. Success means habitual civic action.

2. VERIFIED ACTIVITIES

Audit Precision & Waste Diverted

Number of approved cleanup missions and total metric tons of segregated waste diverted from open streets.

3. OBSERVABLE IMPACT

Cleaner Street Neighborhoods

Longitudinal photographic tracking to verify that cleaned areas remain litter-free over extended periods.

4. SUSTAINABILITY

Partnerships & Operating Trust

Sustaining reward funds through corporate CSR, municipal grants, and preserving user privacy.

SLIDE 6 — CONCLUSION

Small Actions. Collective Impact.

- ✓ Clean & Earn connects community participation with photographic evidence, verification, and incentives.
- ✓ Technology provides structure and anti-fraud verification, while human habits create lasting change.
- ✓ Next step: Launch an on-ground pilot in a college/community to measure empirical results and refine verification.

"A cleaner environment begins when small individual actions become a shared community habit."

Clean & Earn — Encouraging Action, Rewarding Responsibility.

Thank You, Sir! Any Questions? • Live at https://garbage-rosy-six.vercel.app